Sahaj Bhadja

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Summary

Software Engineer with 2+ years building systems software in **C++** and Python, with operating-systems scheduling, concurrency, automated testing, and **troubleshooting**, plus motion-planning research and team delivery with Git and Jira.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built a fault-tolerant **concurrent** job processing system with explicit state transitions and failure isolation, handling 1K+ jobs at 3x throughput without cascading failures under concurrent production load.
- Built integration pipelines with validation and reconciliation checks across SQL databases and cloud storage, **troubleshooting** failures to sustain 99%+ reliability over 5K+ monthly production jobs for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Designed a **multithreaded** batch-processing engine with a thread pool and fault-tolerant execution, processing 1K+ files per run while keeping backend APIs stable for 150+ instructors at peak load.
- Debugged OCR timeouts and model failures in production by adding fallback routing and timeout handling, keeping 10K+ **daily requests** stable and cutting instructor feedback turnaround 60% for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built a **motion planner** for autonomous vehicles that turns 100K+ trajectory logs sampled at 10Hz into discretized control outputs, improving long-horizon trajectory planning 18% in closed-loop simulation across driving scenarios.
- Designed a spatial-temporal encoder fusing map geometry with ego and nearby-agent trajectories, validating it against three baselines in **closed-loop simulation** and improving closed-loop planning stability 15% and reducing off-road trajectories.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Automated nightly builds that compiled the latest code, installed executables on test machines across Windows, Linux, and macOS, and ran 20+ **unit and integration** suites with failure reports.
- Built object-oriented modules with Singleton and Factory patterns, tracked work in **Git and Jira** during Agile sprints, and optimized integrations to improve response times 30% while cutting technical debt 20%.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*
- Designed a quantum-resistant **blockchain** protocol integrating 5+ lattice-based cryptographic primitives and analyzed existing consensus mechanisms to develop a novel consensus algorithm for fair and decentralized validator selection in the network.
- Benchmarked the design across 100 to 1,000 transaction workloads, sustaining 957.8 TPS at 224ms latency, and **presented** the work at IISc Bengaluru as part of an IEEE publication.

Skills

Languages: C++, Python, Java, JavaScript, SQL, Bash, Shell
Systems: Operating Systems, CPU Scheduling, Concurrency, Multithreading, Linux, Windows, Data Structures, Algorithms
Engineering: Object-Oriented Design, Design Patterns, Debugging, Troubleshooting, Simulation, Performance Optimization
Testing & Tools: Unit Testing, Integration Testing, PyTest, Mockito, Git, GitHub, Jira, Jenkins, CI/CD
Other: PyTorch, REST APIs, PostgreSQL, Docker, TCP/IP, Agile, AWS, React, Blockchain, Cryptography

Projects

Pick-and-Place Motion Controller | *C, FreeRTOS, STM32, UART, Unity Tests*
- Designed **FreeRTOS** firmware on an STM32 microcontroller that schedules stepper-motor moves for a small pick-and-place head, reads limit sensors through interrupts, and reports position and fault status over **UART** to a host.

Machine Diagnostics Console | *C++17, Qt, Serial, SQLite, GoogleTest*
- Designed a **C++** desktop tool that talks to equipment controllers over serial, logs sensor and fault data in SQLite, plots cycle times, and helps technicians **diagnose** feeder and nozzle errors.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue **scheduling** in C++ with Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25% across test scenarios.